

The $d = 6$ Exact-Base Branch for E677 Magmas

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Abstract

We isolate and formalize a $d = 6$ exact-base branch in finite E677 magmas. In the branch considered here, a distinguished element x has left orbit $c_0 = x$, $c_1 = x \diamond x$, $c_2 = x \diamond c_1$, ..., $c_5 = x \diamond c_4$, together with an outsider element A satisfying the exact-base rows $A \diamond A = c_1$, $A \diamond x = c_3$, $c_3 \diamond c_3 = c_1$. Under the $d = 6$ gap-1 context hypotheses, these rows force $c_4 \diamond x = x$, and therefore $((x \diamond x) \diamond x) \diamond x = x$. The Lean formalization exposes this result through `D6ExactBasePacket.fixer` in `lean/E677/D6Publication.lean`. The theorem is independent of the large gap-1 dispatcher and verifies with only standard Lean axioms.

1 Background

An E677 magma is a magma $(M, \diamond)$ satisfying the identity

$$a = b \diamond (a \diamond ((b \diamond a) \diamond b))$$

for all $a, b \in M$. We study finite E677 magmas by analyzing left-orbits under maps of the form $L_x(z) = x \diamond z$.

For a fixed non-idempotent element x , write $c_i = L_x^i(x)$. The $d = 6$ branch is the case where the relevant left orbit has six named elements $c_0, c_1, c_2, c_3, c_4, c_5$ with $c_0 = x$ and

$$x \diamond c_i = c_{i+1} \quad \text{for } 0 \leq i < 5, \quad x \diamond c_5 = x.$$

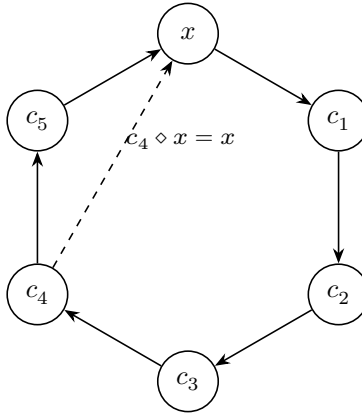


Figure 1: The $d = 6$ orbit of x under L_x . Solid arrows show $x \diamond c_i = c_{i+1}$ (with $c_0 = x$) and $x \diamond c_5 = x$. The dashed arrow is the main conclusion of Theorem 3.1.

The formal Lean context packages the branch as `D6Gap1Context`. This context records the E677 law, the six orbit elements, an outsider A , the $d = 6$ orbit rows, and the auxiliary gap-1 rows needed by the isolated proof.

1.1 The exact-base branch

The larger classifier enumerates all possible behaviors of an outsider element $A \notin \{c_0, \dots, c_5\}$ under the $d = 6$ gap-1 constraints. The exact-base branch is distinguished by three conditions: $A \diamond A$ lands at c_1 (the first non-trivial orbit element), $A \diamond x$ lands at c_3 (the exact midpoint of the orbit, hence “exact base”), and $c_3 \diamond c_3 = c_1$. The last condition is not independent: the gap-1 classifier shows it is forced once the first two hold. Together they constitute the minimal set of rows needed to close the $c_4 \diamond x = x$ derivation.

2 Exact-Base Packet

The exact-base branch adds three rows to the $d = 6$ context:

$$A \diamond A = c_1, \quad A \diamond x = c_3, \quad c_3 \diamond c_3 = c_1.$$

In Lean these rows are packaged as, together with the two publication-facing theorems:

```

structure D6ExactBasePacket (alpha : Type u) [Magma' alpha] [Fintype alpha] where
  ctx : D6Gap1Context alpha
  hAA : ctx.A * ctx.A = ctx.c1
  hA0 : ctx.A * ctx.x = ctx.c3
  hT1 : ctx.c3 * ctx.c3 = ctx.c1

theorem D6ExactBasePacket.fixer (P : D6ExactBasePacket alpha) :
  (((P.ctx.x * P.ctx.x) * P.ctx.x) * P.ctx.x) = P.ctx.x

theorem D6ExactBasePacket.c4_mul_x_eq_x (P : D6ExactBasePacket alpha) :
  P.ctx.c4 * P.ctx.x = P.ctx.x

```

3 Main Theorem

Theorem 3.1. *Let M be a finite E677 magma, and let x be an element in a $d = 6$ gap-1 context with orbit elements $c_0 = x, c_1, c_2, c_3, c_4, c_5$. Suppose there is an outsider A satisfying the exact-base rows*

$$A \diamond A = c_1, \quad A \diamond x = c_3, \quad c_3 \diamond c_3 = c_1.$$

Then

$$((x \diamond x) \diamond x) \diamond x = x.$$

Equivalently, the fourth left iterate of x under L_x returns to x .

4 Proof

The central objective is to establish $c_4 \diamond x = x$. Once this row is known, the conclusion of Theorem 3.1 follows immediately because

$$((x \diamond x) \diamond x) \diamond x = c_4 \diamond x = x.$$

The proof proceeds through three lemmas.

Lemma 4.1 (`exact_packet_c3_mul_c1_eq_x`). $c_3 \diamond c_1 = x$.

Proof. From $A \diamond x = c_3$ and the gap-1 context row $A \diamond (A \diamond x) = A$, substituting gives $A \diamond c_3 = A$. Applying the E677 law with $b = A$, $a = c_3$:

$$\begin{aligned} c_3 &= A \diamond (c_3 \diamond ((A \diamond c_3) \diamond A)) \\ &= A \diamond (c_3 \diamond (A \diamond A)) && \text{(since } A \diamond c_3 = A\text{)} \\ &= A \diamond (c_3 \diamond c_1). && \text{(exact-base: } A \diamond A = c_1\text{)} \end{aligned}$$

Since $A \diamond x = c_3$ as well, left cancellation by A gives $c_3 \diamond c_1 = x$.

Lemma 4.2 (`exact_packet_c1_mul_c4_eq_c3`). $c_1 \diamond c_4 = c_3$.

Proof. Applying the E677 law with $b = c_3$, $a = c_1$:

$$\begin{aligned} c_1 &= c_3 \diamond (c_1 \diamond ((c_3 \diamond c_1) \diamond c_3)) \\ &= c_3 \diamond (c_1 \diamond (x \diamond c_3)) && \text{(Lemma 4.1: } c_3 \diamond c_1 = x\text{)} \\ &= c_3 \diamond (c_1 \diamond c_4). && \text{(orbit: } x \diamond c_3 = c_4\text{)} \end{aligned}$$

Since $c_3 \diamond c_3 = c_1$ (exact-base row), left cancellation by c_3 gives $c_1 \diamond c_4 = c_3$.

Lemma 4.3 (`exact_packet_c4_mul_x_eq_x`). $c_4 \diamond x = x$.

Proof. Applying the E677 law with $b = c_1$, $a = c_4$:

$$\begin{aligned} c_4 &= c_1 \diamond (c_4 \diamond ((c_1 \diamond c_4) \diamond c_1)) \\ &= c_1 \diamond (c_4 \diamond (c_3 \diamond c_1)) && \text{(Lemma 4.2: } c_1 \diamond c_4 = c_3\text{)} \\ &= c_1 \diamond (c_4 \diamond x). && \text{(Lemma 4.1: } c_3 \diamond c_1 = x\text{)} \end{aligned}$$

The $d = 6$ context gives $c_1 \diamond x = c_4$; left cancellation by c_1 gives $c_4 \diamond x = x$.

The orbit definition gives $c_4 = ((x \diamond x) \diamond x) \diamond x$, so Lemma 4.3 completes the proof of Theorem 3.1. The corresponding Lean names are `D6Gap1Context.fixer_of_exact_packet` and `D6ExactBasePacket.fixer`.

5 Formalization Status

The Lean source and paper are available at <https://github.com/mysticflounder/d6-exact-base-e677>.

The standalone publication surface is `lean/E677/D6Publication.lean`, which imports only `E677.D6Isolated`. The isolated proof file `lean/E677/D6Isolated.lean` imports only `E677.D6CoreMinimal`, which in turn imports only `E677.Basic`.

We validated the build with `lake build E677.D6Publication E677.D6TransportPacket E677.D6Full`. Running `lean_verify` on `D6ExactBasePacket.fixer` confirms its axiom footprint is `propext`, `Classical.choice`, `Quot.sound`. The publication theorem does not depend on the remaining project sorry declarations in `Core.lean` or `Gap1.lean`.

6 Acknowledgments

The author acknowledges use of Claude (Anthropic) and Codex/ChatGPT (OpenAI) for proof brainstorming, SAT/DRAT encodings, and code generation for both the Lean proofs and supporting tools.

7 Minimal Extraction Slice

The minimal Lean slice for extraction comprises `lean/E677/Basic.lean`, `lean/E677/D6CoreMinimal.lean`, `lean/E677/D6Isolated.lean`, and `lean/E677/D6Publication.lean`. The large `gap-1` dispatcher remains useful for the full project but is not part of the $d = 6$ exact-base proof body.